

# RESPONSE TO RFI 80JSC027CTTR

## Measurement of Core Temperature and Thermoregulatory Responses During and After Spaceflight in Astronauts

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### 1. Executive Summary

The Instituto de Medicina e Tecnologia Dr. Delyone (IDMT) submits this technical concept in response to NASA RFI 80JSC027CTTR concerning measurement of core temperature and thermoregulatory responses during and after spaceflight.

The proposed approach is not limited to developing a more accurate thermometer. It is intended to characterize thermoregulation as a coupled physiological and environmental process, integrating deep-body temperature, vascular-proximal temperature, peripheral temperature, exercise/metabolic load, and the thermal conditions surrounding the crewmember.

The central premise is:

**Core temperature is a physiological variable; thermal stability is a system property.**

We propose a multimodal measurement architecture that could combine:

1. a minimally invasive, biocompatible deep thermal reference;
2. paired temperature measurements positioned in proximity to arterial and venous structures, without intravascular placement;
3. simultaneous non-invasive surface/heat-flux measurements;
4. skin and environmental thermal measurements;
5. controlled exercise and other physiological perturbations; and
6. longitudinal measurements spanning preflight, inflight, and postflight periods.

The concept includes a proposed approximately 60-day preflight period for device validation, individual baseline characterization, and comparison with established non-invasive measurements before launch. The same measurement architecture could then be used longitudinally through spaceflight and recovery, subject to medical, ethical, engineering, and flight-qualification requirements.

This approach is intended to help distinguish whether observed changes in measured core temperature are associated primarily with changes in metabolic heat production, cardiovascular/tissue transport, peripheral heat dissipation, environmental boundary conditions, or interactions among these factors.

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### 2. Understanding of the RFI

The RFI seeks information relevant to characterizing core body temperature and thermoregulatory responses in astronauts before, during, and after spaceflight, including rest, nominal operations, exercise, and re-entry/landing.

The RFI identifies esophageal and rectal measurements as accepted reference approaches while noting their limitations for routine spaceflight. It also identifies the non-invasive Double Sensor heat-flux approach as promising, while noting that it has not been validated against accepted core-temperature measures during actual spaceflight.

The RFI further identifies the operational importance of accurate thermal characterization for thermal modeling, physical and cognitive performance, thermal injury risk, orthostatic responses, thermal-control hardware, environmental control, and operational constraints.

The proposed IDMT framework is therefore designed around two complementary objectives:

- improve the physiological reference measurement available for validation; and
  - characterize the mechanisms and boundary conditions that determine the observed thermal response.
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### **3. Scientific Rationale**

Published ISS observations have demonstrated that thermoregulatory behavior during long-duration spaceflight may differ materially from terrestrial baseline.

Stahn et al. (2017), studying 11 astronauts during approximately 180-day ISS missions, reported an approximately 1.01 °C increase in estimated resting core body temperature during spaceflight relative to preflight baseline, developing progressively over approximately the first 2.5 months and reaching a plateau after approximately day 75.

During exercise, maximum estimated core temperature was approximately 1.39 °C higher in flight, and the reported rate of core-temperature increase was approximately 0.10 °C/min on Earth compared with approximately 0.15 °C/min during spaceflight.

The measurements used a non-invasive Double Sensor heat-flux method involving two temperature probes separated by an insulating element. The method estimates core temperature from surface temperature and thermal-gradient/heat-flux information rather than directly measuring a deep anatomical temperature.

These observations demonstrate the value of continuous non-invasive monitoring, but they also highlight the need for an independent reference capable of determining how accurately peripheral measurements represent deep-body thermal state under changing spaceflight conditions.

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## **4. Proposed Measurement Architecture**

### **4.1 Deep Thermal Reference**

We propose investigation of a miniature, biocompatible, minimally invasive implantable temperature sensor intended to provide a stable deep-tissue thermal reference.

The concept is deliberately non-intravascular. The sensor would be positioned in a tissue location selected through anatomical imaging, safety analysis, preclinical assessment, and human-subject evaluation.

The final anatomical site should be determined by the requirements of measurement fidelity, vascular anatomy, implantation safety, biocompatibility, crew acceptability, and flight operations.

### **4.2 Paired Arterial/Venous-Proximal Measurements**

A particularly novel element of the concept is the investigation of two thermally isolated measurement points positioned in proximity to paired arterial and venous structures.

The objective is not to assume that venous blood is simply “colder” than arterial blood. Rather, the paired measurements would provide temporally resolved observations of thermal transport through a defined vascular/tissue territory.

Potential variables include:

- arterial-proximal temperature,  $T_a(t)$ ;
- venous-proximal temperature,  $T_v(t)$ ;
- local arterial-venous temperature difference,  $\Delta T_{AV} = T_a - T_v$ ;
- $dT_a/dt$ ;
- $dT_v/dt$ ;
- $d\Delta T_{AV}/dt$ ;
- local tissue temperature;
- surface/skin temperature;
- environmental temperature and humidity; and
- exercise/metabolic workload.

These variables could be analyzed jointly to investigate changes in thermal production, transport, and dissipation.

Importantly,  $\Delta T_{AV}$  should be treated as a complementary physiological signal rather than a direct measurement of total body heat loss. Its interpretation would require calibration against accepted reference measurements and consideration of local blood flow, tissue perfusion, metabolism, and environmental conditions.

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## 5. Simultaneous Non-Invasive Measurement

The proposed deep reference should not replace non-invasive monitoring. Instead, the two approaches should be deliberately coupled.

During validation, the implantable reference could be measured simultaneously with:

- Double Sensor or comparable heat-flux technology;
- skin temperature;
- heart rate and cardiovascular variables;
- exercise workload;
- environmental temperature;
- relative humidity;
- airflow/ventilation conditions;
- hydration and relevant behavioral factors; and
- other available physiological measurements.

This would allow the non-invasive signal to be evaluated under controlled changes in thermal boundary conditions rather than only under static laboratory conditions.

A central question would be:

When a non-invasive estimate of core temperature changes, does the change correspond to a true change in deep-body thermal state, or does the relationship between the peripheral measurement and deep temperature change because the thermal boundary conditions have changed?

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## 6. Proposed 60-Day Preflight Characterization

We propose an approximately 60-day preflight period between implantation/activation and launch, subject to medical and operational feasibility.

The purpose would be to establish an individual physiological and measurement baseline and to verify device stability before flight.

Potential objectives include:

- confirmation of sensor stability;
- assessment of signal quality and wireless communication;
- evaluation of local tissue response and biocompatibility;
- assessment for migration or other implantation-related complications;
- comparison with accepted/non-invasive reference measurements;
- characterization during rest and sleep;
- characterization during controlled exercise;
- assessment across ordinary daily activity;
- assessment under controlled environmental conditions; and
- establishment of individual thermal-response signatures.

The proposed period also creates an opportunity to identify unacceptable device behavior before launch. If medical or measurement concerns arise, the device could be evaluated for removal or other appropriate management before flight, subject to the applicable clinical protocol.

The 60-day interval is a proposed research design element, not an assertion of an established flight protocol.

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## 7. Exercise as a Controlled Physiological Perturbation

Exercise provides a particularly useful experimental condition because metabolic heat production increases rapidly and predictably relative to rest.

During exercise, active skeletal muscle increases ATP turnover and metabolic energy expenditure, with a substantial fraction of the chemical energy ultimately appearing as heat.

Accordingly, the experiment can examine not only absolute temperature, but also the dynamics of temperature change.

Potential analytical variables include:

- baseline temperature;
- peak temperature;
- $dT/dt$  during exercise;
- recovery slope;
- time to peak;
- arterial-proximal/venous-proximal temperature relationships;
- skin-to-deep temperature gradients;
- environmental conditions; and
- workload-normalized thermal response.

The same exercise protocol could be performed preflight, inflight, and postflight where operationally feasible.

This would create a longitudinal perturbation-response dataset rather than a series of isolated temperature measurements.

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## 8. Human Thermal Regulation as a Coupled System

The proposed framework treats human thermal regulation as a component of a larger coupled system.

A simplified hierarchy is:

**External thermal environment** → **spacecraft thermal-control system** → **cabin environment** → **human/environment interface** → **human physiology**

The human component includes:

- metabolic heat production;
- cardiovascular transport;
- tissue heat storage;
- skin blood flow;
- sweating and evaporation;
- radiation;
- convection;
- conduction; and
- behavioral responses.

The spacecraft environment includes:

- ventilation and forced airflow;
- cabin temperature;
- humidity;
- thermal gradients;
- heat-transfer pathways;
- radiative conditions;
- environmental-control systems; and
- thermal-control hardware.

This distinction is important because a measured change in body temperature does not, by itself, identify which component of the coupled system changed.

The proposed architecture therefore seeks to measure the human thermal state while simultaneously characterizing the thermal boundary conditions in which that state exists.

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## 9. Inflight and Postflight Continuity

Where flight qualification and mission operations permit, the same measurement architecture could be maintained across:

**Preflight** → **Inflight** → **Re-entry/Landing** → **Recovery**

This continuity would reduce uncertainty introduced by switching measurement modalities at different phases of the mission.

Potential postflight assessments could examine:

- return toward individual baseline;
- persistence of altered thermal responses;
- recovery of exercise thermoregulation;
- changes in peripheral/deep temperature relationships; and
- relationships between thermal response and orthostatic or other physiological recovery.

The same framework could ultimately be investigated across different mission durations and environments, including LEO, lunar missions, Mars missions, and relevant terrestrial analogs.

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## 10. Wireless and Implantable Implementation

A future implementation could investigate passive or low-power wireless telemetry, including NFC-derived architectures where appropriate.

Such implementation would require dedicated engineering assessment of:

- communication depth;
- antenna geometry;
- power requirements;
- electromagnetic compatibility;
- sterilization;
- biocompatibility;
- mechanical robustness;
- thermal coupling;
- data integrity;
- radiation environment;
- spacecraft integration; and
- crew safety.

The purpose of identifying wireless/NFC architectures at this stage is to describe a potential development pathway, not to imply that a flight-qualified system already exists.

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## 11. Validation Strategy

A central requirement of the proposed research is validation against accepted physiological reference measurements.

The development sequence could include:

### Phase 1 — Bench and thermal-phantom testing

Characterize sensor response, thermal time constants, drift, calibration, wireless performance, and sensitivity to controlled thermal gradients.

### Phase 2 — Preclinical safety and feasibility

Evaluate implantation approach, tissue response, device stability, thermal coupling, and retrieval/removal considerations.

### Phase 3 — Human terrestrial validation

Compare the implantable measurements with accepted core-temperature reference methods under controlled rest, exercise, thermal stress, hydration, and environmental conditions.

## **Phase 4 — Analog environments**

Evaluate the system under bed-rest and other relevant operational analogs to separate environmental, inactivity, and microgravity-associated effects.

## **Phase 5 — Flight validation**

If prior phases demonstrate acceptable safety, accuracy, and operational feasibility, conduct controlled astronaut measurements before, during, and after spaceflight.

This staged approach is intended to reduce the risk of introducing an unvalidated measurement system into an operational flight environment.

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# **12. Key Scientific Questions**

The proposed framework is designed to address several questions directly relevant to the RFI:

1. Does the approximately 1 °C increase in estimated resting core temperature observed during long-duration spaceflight represent a true change in deep-body temperature, a change in thermoregulatory set point/equilibrium, a measurement relationship effect, or a combination?
  2. How does the relationship between deep-body temperature and non-invasive surface/heat-flux estimates change in the spaceflight environment?
  3. Does exercise produce a different temporal relationship between metabolic load, vascular thermal transport, peripheral temperature, and deep-body temperature in spaceflight?
  4. Can simultaneous vascular-proximal measurements provide additional information about thermal transport through tissue?
  5. Which environmental variables most strongly influence the relationship between peripheral measurements and deep thermal state?
  6. Can a multimodal thermal dataset improve physiological inputs to spacecraft and spacesuit thermal models?
  7. Which measurements are sufficiently robust and operationally simple for long-duration missions beyond LEO?
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# **13. Expected Contribution**

The proposed contribution is a measurement framework rather than a single replacement sensor.

The intended outcome is a better-resolved model of the astronaut thermal state that links:

**metabolic heat production → cardiovascular/tissue transport → tissue heat storage → peripheral dissipation → environmental boundary conditions → measured core temperature**

Such a framework could provide NASA with data useful for improving thermal models, interpreting thermoregulatory responses, validating non-invasive monitoring technologies, and evaluating future thermal countermeasures.

The proposed architecture is particularly intended to support developmental pathways for constrained missions where invasive reference measurements are impractical but high-confidence physiological characterization remains important.

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## 14. Developmental Status and Limitations

The implantable paired vascular-proximal architecture described in this response should be considered a research concept requiring engineering development and validation.

The proposal does not claim that:

- the proposed implant has already been flight qualified;
- the proposed anatomical placement has already been established as safe for astronauts;
- the arterial-venous temperature difference is by itself a validated measure of whole-body heat loss; or
- the proposed wireless implementation is already suitable for ISS, lunar, or Mars operations.

These are precisely the questions that a staged development and validation program would address.

The concept should be evaluated through appropriate institutional review, human-subject protections, medical safety assessment, biocompatibility testing, device qualification, electromagnetic compatibility testing, and mission-specific engineering review.

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## 15. Conclusion

The RFI identifies an important measurement gap: accepted deep-body reference methods are difficult to use routinely during spaceflight, while non-invasive methods provide operational advantages but require stronger validation under actual spaceflight conditions.

The IDMT proposes addressing this gap through a multimodal architecture that combines a minimally invasive deep thermal reference with non-invasive monitoring and simultaneous characterization of physiological and environmental variables.

The central scientific objective is not simply to determine whether an astronaut is “37 °C.”

It is to determine **why the astronaut’s thermal state changes, how that change propagates through the body, how the environment modifies the response, and whether the measured signal represents physiology, boundary conditions, or both.**

**Core temperature is a physiological variable; thermal stability is a system property.**

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## 16. Organization

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**This document does not constitute an offer, proposal, or commitment to perform work.**

